

Problem-Solution Fit Canvas

Date	15 July 2026
Team ID	XXXXXX
Project Name	FloodGuard - AI/ML Flood Prediction System
Maximum Marks	5 Marks

1. CUSTOMER SEGMENT(S) Local residents in flood-prone areas; disaster-management / municipal officers; agriculture and civic planning teams	6. CUSTOMER LIMITATIONS Limited access to specialized meteorology tools; rely on generic weather apps; low tolerance for complex dashboards	5. AVAILABLE SOLUTIONS (PROS & CONS) Govt. weather bulletins: authoritative but not localized/predictive. News alerts: reactive, not risk-scored. Manual expert judgment: accurate but slow and not scalable
2. PROBLEMS / PAINS + FREQUENCY No simple flood/no-flood signal from raw rainfall numbers — happens every monsoon season, repeatedly	9. PROBLEM ROOT / CAUSE Weather data (rainfall, humidity, temperature, cloud cover) is not translated into an actionable risk score for non-experts	7. BEHAVIOR + INTENSITY Users check weather apps repeatedly and anxiously during heavy rain, but still hesitate to act until visible flooding starts (high intensity, low confidence)
3. TRIGGERS TO ACT Sudden heavy rainfall alerts; visible water-level rise near rivers/drains	10. YOUR SOLUTION FloodGuard: an ML-powered web app that converts live weather inputs into an instant flood-risk probability with a clear Safe / Flood Risk badge	8. CHANNELS OF BEHAVIOR Online: weather apps, news websites, WhatsApp alerts. Offline: word-of-mouth from neighbours, local announcements